Gradient Compensation Enabled Federated Learning for Unreliable Wireless Links

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***Abstract*—Wireless federated learning (FL) faces significant challenges due to limited wireless resources and unreliable channels. To cope with these challenges, this work proposes a gradient compensation-based FL approach (FL-GC), in which the edge server estimates the local gradients of transmission failure and unselected clients by first-order Taylor approximation based on previously received local gradients. We then theoretically analyze the convergence bound, which reveals that selecting clients with large local gradient staleness helps reduce the estimation error and improve learning performance. Based on this, we jointly optimize the client selection and resource allocation strategies to enhance the FL performance under resource-limited wireless networks. Simulation results under a typical data heterogeneity scenario demonstrate the efficacy of our proposed scheme in mitigating the adverse effects of unreliable transmission and limited resources. It improves 7.34% model accuracy compared to the considered benchmarks and is able to save 42.5% training time to achieve the target accuracy.**

***Index Terms*—Federated Learning, gradient compensation, unreliable communication**

I. Introduction

Federated learning (FL) is a promising machine learning paradigm that exploits massive data at the wireless edges to promote edge intelligence services (e.g., autonomous driving) while preserving users' privacy. However, the deployment of FL in wireless networks confronts the limited wireless resources, which restricts the participation client number in the training process and thus degrades the learning performance, especially under data heterogeneous scenarios [1]. Furthermore, the inherently unreliable nature of wireless channels poses a significant challenge to clients' successful transmission of local models, thereby exacerbating the performance deterioration for FL. Therefore, it is crucial to design efficient FL frameworks to tackle the above two challenges.

To tackle the limited wireless resources, existing works focused on optimizing the client selection strategy design to improve the learning performance of FL. Specifically, the channel condition-aware client selection approach in [2] effectively improved the learning accuracy and reduced the energy consumption of FL. A channel condition and gradient importance-based client selection approach has been developed in [3] to accelerate FL convergence. The joint client representativity and latency-aware client selection scheme in [4] effectively tackled the data heterogeneity issues for resource-limited wireless FL. While the client selection schemes effectively alleviate the resource-limited bottlenecks, the unreliable nature of wireless channels has not been investigated.

To address the challenges posed by unreliable communication, existing works concentrated on maximizing the successful transmission probabilities of clients [5], designing retransmission mechanisms [6], [7], and reusing past models for clients that encounter transmission failures [8], [9]. Specifically, the client selection and resource allocation approach in [5] significantly improves the number of successful transmission cleints. Compared to the successful transmission probabilities maximization approaches, the retransmission protocols in [6] and [7] further diminished transmission errors in wireless FL. In the retransmission strategies, each client undertakes multiple transmissions of its local model, and the edge server (ES) recovers the local model from the received signal featuring the highest signal-to-noise ratio (SNR). However, frequent retransmission may cause additional learning costs. In addition, the above successful transmission probabilities maximization and retransmission approaches only aggregate successfully transmitted local models. The absence of updates from the transmission failure clients still leads to biased model aggregation and degrades FL performance. To address this issue, the model compensation approaches in [8] and [9] were developed to accelerate the FL convergence by replacing error models with past local models in case of clients' model uploading errors. However, the gap between past and real current local models may introduce errors in global aggregation, especially when clients' past models are too stale.

According to the above discussion, this work jointly designs the wireless network and learning mechanism to mitigate the adverse impact of unreliable wireless channels and limited resources for FL. Specifically, we propose a new gradient compensation-based FL approach, i.e., **FL-GC**,

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that estimates the local gradients of transmission failure and unscheduled clients based on the latest historical gradients received at the ES for global model updating. Compared to the past model/gradient compensation approaches in [8]–[10], the proposed FL-GC further reduces the global model aggregation errors and obtains better learning performance. We theoretically analyze the convergence bound of FL-GC, which shows that minimizing the average staleness of local gradients helps reduce the estimation errors and improve the learning performance. Inspired by this, we develop a joint optimal client selection and resource allocation approach to improve the FL performance under resource-constrained wireless networks. Simulation results show that the proposed approach improves 7.34% accuracy compared to the considered benchmarks.

II. System Model

This work considers a typical wireless FL system, where N clients and an ES cooperatively train a machine learning model. The clients are indexed by $\mathcal{N} = \{1, 2, \cdots, N\}$. Each client n ($n \in \mathcal{N}$) possesses a local dataset $\mathcal{D}_n$ with D_n data samples. The entire dataset is $\mathcal{D} = \cup_{n=1}^{N} \mathcal{D}_n$ with $D = \sum_{n=1}^{N} D_n$ samples. For each data sample $\zeta \in \mathcal{D}$, let $l(\zeta; \mathbf{x})$ be the sample-wise loss function. Thus, the local loss function of client n ($n \in \mathcal{N}$) is

$$\mathcal{L}_n(\mathbf{x}) = \frac{1}{D_n} \sum\nolimits_{\zeta \in \mathcal{D}_n} l(\zeta; \mathbf{x}). \tag{1}$$

The goal of FL is to train a global shared model $\mathbf{x}$ for minimizing the global loss function as

$$\mathcal{L}(\mathbf{x}) = \frac{1}{N} \sum\nolimits_{n=1}^{N} \mathcal{L}_n(\mathbf{x}). \tag{2}$$

A. Federated Learning with Gradient Compensation

This work proposes FL-GC to mitigate the adverse effects of constrained resources and unreliable transmission on FL, in which the ES applies the *Taylor approximation* to estimate the gradients of transmission failure and unselected clients for global model updating. FL-GC performs T global rounds of training, and each round t consists of four steps as follows:

1) *Global Model Broadcasting*: The ES chooses a set of clients and broadcasts the latest global model to them for local training. Let $\theta_{n,t} \in \{0, 1\}$ be the client selection decision, $\theta_{n,t} = 1$ if n is selected in round t, and $\theta_{n,t} = 0$ otherwise.

2) *Local Training*: Each selected client n performs I-steps stochastic gradient descent to update its local model as:

$$\mathbf{x}_{n,t}^{(i+1)} = \mathbf{x}_{n,t}^{(i)} - \eta \tilde{\nabla} \mathcal{L}_n(\mathbf{x}_{n,t}^{(i)}), i = 0, \cdots, I-1, \tag{3}$$

where $\mathbf{x}_{n,t}^{(i)}$ is client n's model in the i-th iteration of round t, $\mathbf{x}_{n,t}^{(0)} = \mathbf{x}_t$, and $\eta > 0$ is the learning rate. In (3), the stochastic gradient $\tilde{\nabla} \mathcal{L}_n(\mathbf{x}_{n,t}^{(i)})$ is computed as

$$\tilde{\nabla} \mathcal{L}_n(\mathbf{x}_{n,t}^{(i)}) = \frac{1}{H_b} \sum\nolimits_{\zeta \in \mathcal{H}_{n,t,i}} \nabla l(\zeta; \mathbf{x}_{n,t}^{(i)}), \tag{4}$$

where $\mathcal{H}_{n,t,i}$ is a mini-batch data uniformly selected from $\mathcal{D}_n$ with $H_b = |\mathcal{H}_{n,t,i}|$ data samples.

3) *Local Gradient Uploading*: Each selected client n uploads its gradient $\tilde{\mathbf{g}}_{n,t}$ to the ES. $\tilde{\mathbf{g}}_{n,t}$ is expressed as

$$\tilde{\mathbf{g}}_{n,t} = \sum\nolimits_{i=0}^{I-1} \tilde{\nabla} \mathcal{L}_n(\mathbf{x}_{n,t}^{(i)}) = \frac{1}{\eta}(\mathbf{x}_t - \mathbf{x}_{n,t}^{(I)}). \tag{5}$$

Due to unreliable wireless channels, the local gradients may not be successfully transmitted to the ES. Let $s_{n,t} \in \{0, 1\}$ be the successful transmission indicator of client n in round t, where $s_{n,t} = 1$ represents that its local gradient is successfully transmitted, and $s_{n,t} = 0$ otherwise.

4) *Global Model Update*: The ES updates the global model based on the aggregated local gradients from clients. Unlike existing works, e.g., [5], [6], that directly disregard the unsuccessfully received local gradients. This work adopts the Taylor approximation to estimate un-received clients' local gradients based on their latest received gradients at the ES. Then, the ES aggregates the successfully received gradients and the estimated gradients as:

$$\widehat{\mathbf{g}}_t = \frac{1}{N} \sum_{n=1}^{N} (\theta_{n,t} s_{n,t} \tilde{\mathbf{g}}_{n,t} + (1 - \theta_{n,t} s_{n,t}) \widehat{\mathbf{g}}_{n,t}) \tag{6}$$

where $\widehat{\mathbf{g}}_{n,t}$ is estimated gradient of client n. Finally, the ES updates the global model as $\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \widehat{\mathbf{g}}_t$.

In (6), the Taylor expansion of the estimated gradient of client n is given by

$$\begin{aligned} \widehat{\mathbf{g}}_{n,t} = \tilde{\mathbf{g}}_{n,t-\tau} + \nabla \tilde{\mathbf{g}}_{n,t-\tau}(\mathbf{x}_t - \mathbf{x}_{t-\tau}) \\ + O\left((\mathbf{x}_t - \mathbf{x}_{t-\tau})^2\right) I_n, \end{aligned} \tag{7}$$

where $\tilde{\mathbf{g}}_{n,t-\tau}$ is the previous successfully received gradient from client n at the ES, $\nabla \tilde{\mathbf{g}}_{n,t-\tau}$ is the Hessian matrix of its loss function, I_n is a n-dimension vector with all elements equal to 1. An ideal method is to utilize the full Taylor expansion to estimate the local gradients. However, this is not practical since it involves the sum of an infinite number of terms.

To tackle this issue, we adopt the first-order Taylor approximation to estimate the gradient of client n, i.e.,

$$\widehat{\mathbf{g}}_{n,t} \approx \tilde{\mathbf{g}}_{n,t-\tau} + \nabla \tilde{\mathbf{g}}_{n,t-\tau}(\mathbf{x}_t - \mathbf{x}_{t-\tau}). \tag{8}$$

However, the first-order approximation is still computationally expensive since it requires the computation of the Hessian matrix. According to [11], the outer product of the local gradient is an asymptotically unbiased estimation of the Hessian matrix. Thus, this work adopts the outer product matrix of local gradients to approximate the Hessian matrix as $\nabla \tilde{\mathbf{g}}_{n,t-\tau} = \tilde{\mathbf{g}}_{n,t-\tau} \tilde{\mathbf{g}}_{n,t-\tau}^T$. Thus, the estimation of client

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n's gradient is given by

$$\widehat{\mathbf{g}}_{n,t} \approx \tilde{\mathbf{g}}_{n,t-\tau} + \tilde{\mathbf{g}}_{n,t-\tau}\tilde{\mathbf{g}}_{n,t-\tau}^{T}(\mathbf{x}_t - \mathbf{x}_{t-\tau}). \tag{9}$$

B. Successful Transmission Model

In this work, the orthogonal frequency division multiple access with C resource block groups (RBGs) indexed by $\mathcal{C} = \{1, 2, \cdots, C\}$ is utilized for clients to upload their gradients. Let $\boldsymbol{z}_{n,t} = (z_{n,t}^{(1)}, z_{n,t}^{(2)}, \cdots, z_{n,t}^{(C)})$ denote the RBG allocation vector for client n in round t, where $z_{n,t}^{(c)} \in \{0, 1\}$, $z_{n,t}^{(c)} = 1$ represents that the c-th RBG is allocated to client n, and $z_{n,t}^{(c)} = 0$ otherwise. It is noted that each client can only occupy at most one RBG, and each RBG can be accessed by at most one client. Thus, the RBG allocation policy should satisfy $\sum_{c=1}^{C} z_{n,t}^{(c)} \leq 1$ and $\sum_{n=1}^{N} z_{n,t}^{(c)} \leq 1$. Let p_n be the transmit power of client n. The channel gain of client n is modelled as $h_{n,t} = \rho_n(t) d_n^{-v}$, where $\rho_n(t)$ is the small-scale fading gain between client n and the ES, d_n is the distance from client n to the ES, and v is the path loss factor. This work considers Rayleigh fading, i.e., $\rho_n(t) \sim \exp(1)$. Thus, the SNR of client n in the c-th RBG is $\gamma_{n,t}^{(c)} = \frac{p_n h_{n,t}}{B N_0}$, where B is the bandwidth of each RBG and N_0 is the noise power spectral density.

Let γ_{th} be the SNR threshold for successful data decoding. Thus, $s_{n,t} = \sum_{c=1}^{C} z_{n,t}^{(c)} \mathbb{1}(\gamma_{n,t}^{(c)} \geq \gamma_{\text{th}})$ $(\forall n \in \mathcal{N})$. The probability of error-free transmission for client n through c-th RBG is

$$\begin{aligned} \Pr(\gamma_{n,t}^{(c)} \geq \gamma_{\text{th}}) &= \Pr\left(\rho_n(t) \geq \frac{\gamma_{\text{th}} B N_0}{p_n d_n^{-v}}\right) \\ &= \exp\left(-\frac{\gamma_{\text{th}} B N_0}{p_n d_n^{-v}}\right). \end{aligned} \tag{10}$$

C. Learning Latency Model

This subsection characterizes the per-round latency of FL.

1) *Computation Latency*: Let C_n denotes the required CPU cycles to process one data sample at client n and f_n denote the CPU frequency of client n. Thus, the computation time is

$$T_{n,t}^{\text{C}} = \frac{I H_b C_n}{f_n}. \tag{11}$$

2) *Communication Latency*: According to the above discussed communication model, the transmit rate of client n in round t is

$$r_{n,t}(\boldsymbol{z}_{n,t}) = \sum\nolimits_{c=1}^{C} z_{n,t}^{(c)} B \log_2(1 + \gamma_{n,t}^{(c)}). \tag{12}$$

Let R denote the data size of each gradient. The communication latency of client n in round t is

$$T_{n,t}^{\text{U}} = \frac{R}{r_{n,t}(\boldsymbol{z}_{n,t})}. \tag{13}$$

It is worth mentioning that we disregard the latency of the global model broadcast and update. This is because the ES has large transmit power and powerful computation capabilities, which results in the broadcasting and updating latency being negligible compared to the aforementioned latencies.

III. Convergence Analysis and Optimization

A. Convergence Analysis

Let $\mathcal{L}(\mathbf{x}^*)$ denote the loss function of the optimal global model $\mathbf{x}^*$. Note that FL-GC estimates the un-received local gradients based on clients' previously successfully received local gradients. We introduce a variable, i.e., $\tau_{n,t}$, to characterize the staleness of the previously received gradient of client n. $\tau_{n,t}$ is given by:

$$\tau_{n,t} = \begin{cases} \tau_{n,t-1} + 1, & \text{if } \theta_{n,t} s_{n,t} = 0, \\ 0, & \text{if } \theta_{n,t} s_{n,t} = 1. \end{cases} \forall n \in \mathcal{N}. \tag{14}$$

To begin with, we make some standard assumptions on the loss functions as follows [4], [5], [8]:

Assumption 1. All the local loss functions, $\mathcal{L}_n(\mathbf{x})$ $(\forall n \in \mathcal{N})$, are L-smooth. That is, for all $\mathbf{x}_1$ and $\mathbf{x}_2$,

$$\|\nabla \mathcal{L}_n(\mathbf{x}_1) - \nabla \mathcal{L}_n(\mathbf{x}_2)\| \leq L \|\mathbf{x}_1 - \mathbf{x}_2\|. \tag{15}$$

Assumption 2. The stochastic gradient $\tilde{\nabla}\mathcal{L}_n(\mathbf{x}_t)$ is an unbiased estimation of the full gradient $\nabla \mathcal{L}_n(\mathbf{x}_t)$, i.e., $\mathbb{E}[\tilde{\nabla}\mathcal{L}_n(\mathbf{x}_t)] = \nabla \mathcal{L}_n(\mathbf{x}_t)$, and its variance satisfies $\mathbb{E}\|\tilde{\nabla}\mathcal{L}_n(\mathbf{x}_t) - \nabla \mathcal{L}_n(\mathbf{x}_t)\|^2 \leq \sigma^2$.

Assumption 3. The squared norm of local gradients and the estimated gradients are bounded by G_1^2 and G_2^2, respectively, i.e., $\|\nabla \mathcal{L}(\mathbf{x}_t)\|^2 \leq G_1^2$ and $\|\widehat{\mathbf{g}}\|^2 \leq G_2^2$. The second order gradients of clients are bounded by ε, i.e., $||\nabla \tilde{\mathbf{g}}_{k,t}||^2 \leq \varepsilon$.

In the following, we first introduce one lemma to facilitate our convergence analysis, proved in Appendix A.

Lemma 1. *If Assumptions* 1-3 hold and $\eta \leq \frac{1}{2IL}$, the global model drift between round t and t' $(t > t')$ is bounded by

$$\|\mathbf{x}_t - \mathbf{x}_{t'}\|^2 \leq 6\eta^2 c_1 (t - t')^2, \tag{16}$$

where $c_1 = \frac{I^2 + I - 1}{I^2}\sigma^2 + \frac{3I-2}{I} G_1^2 + \frac{G_2^2}{3}$.

Building upon Lemma 1, we present the convergence bound of FL-GC in the following theorem, proved in Appendix B.

Theorem 1. *If Assumptions* 1-3 hold and $\eta \leq \frac{1}{2IL}$, the convergence bound of FL-GC is:

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$$\mathbb{E}[\mathcal{L}(\mathbf{x}_{t+1}) - \mathcal{L}(\mathbf{x}^*)] \leq (1 - L^2\eta^2)^{T-1}\mathbb{E}[\mathcal{L}(\mathbf{x}_0) - \mathcal{L}(\mathbf{x}^*)] + c + 3\eta^4 L\varepsilon c_1 \sum_{t=1}^{T-1}(1 - L^2\eta^2)^t \frac{1}{N}\sum_{n=1}^{N}(\tau_{n,t-1}+1)^2 \times \left(1 - \theta_{n,t}\sum_{c=1}^{C} z_{n,t}^{(c)} \exp\left(-\frac{\gamma_{\text{th}} B N_0}{p_n d_n^{-v}}\right)\right), \quad (17)$$

where $c = \frac{1-L^2\eta^2-(1-L^2\eta^2)^T}{L}((2I^2-2I+1)\sigma^2+(I-1)^2G_1^2)$.

From Theorem 1, the convergence bound of FL-GC is determined by three terms: 1) The initial gap. 2) A constant term decided by system hyperparameters. 3) A wireless network parameters-related term. In practical wireless networks, one can optimize the third term on the right-hand side (RHS) of (17) to improve the learning performance of FL-GC. To this end, we have the following remark:

Remark 1. According to Theorem 1, we define a new objective function, i.e., $\frac{1}{N}\sum_{n=1}^{N}(\tau_{n,t-1}+1)^2\theta_{n,t}\sum_{c=1}^{C} z_{n,t}^{(c)}\exp(-\frac{\gamma_{\text{th}} B N_0}{p_n d_n^{-v}})$. Maximizing this objective function in each round achieves the convergence bound minimization. Note that the client selection policy satisfy $\theta_{n,t} = \sum_{c=1}^{C} z_{n,t}^{(c)}$. Thus, the objective function can be rewritten as

$$h(\boldsymbol{Z}_t) = \frac{1}{N}\sum_{n=1}^{N}\sum_{c=1}^{C} z_{n,t}^{(c)}(\tau_{n,t-1}+1)^2 \exp\left(-\frac{\gamma_{\text{th}} B N_0}{p_n d_n^{-v}}\right), \quad (18)$$

where $\boldsymbol{Z}_t = (\boldsymbol{z}_{1,t}, \boldsymbol{z}_{2,t}, \cdots, \boldsymbol{z}_{N,t})$.

B. Optimization of Client Selection and Resource Allocation

Drawing from the insights of Remark 1, we focus on maximizing $h(\boldsymbol{Z}_t)$ in each round by optimizing the resource allocation policies. The formulation of the optimization problem is:

$$\mathcal{P}: \max_{\boldsymbol{Z}_t} \ h(\boldsymbol{Z}_t) \quad (19)$$

$$\text{s. t. } T_{n,t}^{\text{C}} + T_{n,t}^{\text{U}} \leq T_{\max}, \forall n \in \mathcal{N}, \forall t, \quad (19a)$$

$$z_{n,t}^{(c)} \in \{0,1\}, \forall n \in \mathcal{N}, \forall c \in \mathcal{C}, \forall t, \quad (19b)$$

$$\sum_{c=1}^{C} z_{n,t}^{(c)} \leq 1, \forall n \in \mathcal{N}, \forall t, \quad (19c)$$

$$\sum_{n=1}^{N} z_{n,t}^{(c)} \leq 1, \forall c \in \mathcal{C}, \forall t. \quad (19d)$$

In problem $\mathcal{P}$, (19a) stipulates that the per-round latency cannot exceed the maximum allowable latency $T_{\max}$ whilst (19b)-(19d) correspond to the RBG allocation constraints.

Problem $\mathcal{P}$ is a typical nonlinear integer programming that is renowned for its challenges in finding a solution. Below we transform it into a maximum weight bipartite matching problem and find its optimal solution within polynomial time. To this end, we construct a bipartite graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$, where $\mathcal{V} = \mathcal{N} \cup \mathcal{C}$ is the vertex set, and $\mathcal{E}$ is the set of edges that connect the vertices in $\mathcal{N}$ and $\mathcal{C}$. In $\mathcal{G}$, each vertex n in $\mathcal{N}$ corresponds a client n, and each vertex c in $\mathcal{C}$ corresponds to RBG c. The weight of edges in $\mathcal{G}$ is given by:

$$\Delta_{n,c} = \begin{cases} (\tau_{n,t-1}+1)^2 \exp\left(-\frac{\gamma_{\text{th}} B N_0}{p_n d_n^{-v}}\right), & \text{if } T_{n,t}^{\text{C}} + T_{n,t}^{\text{U}} \leq T_{\max}, \\ 0, & \text{otherwise.} \end{cases} \quad (20)$$

According to the bipartite graph $\mathcal{G}$, we transform $\mathcal{P}$ into a maximum weight bipartite matching problem, which aims to find a matching $\mathcal{S}$ of $\mathcal{G}$ maximizing $\sum_{e \in \mathcal{S}} \Delta_{n,c}$. A standard Kuhn-Munkres algorithm [12] can be deployed to find the optimal solution with the time complexity of $\mathcal{O}(N^3)$.

IV. Numerical Results

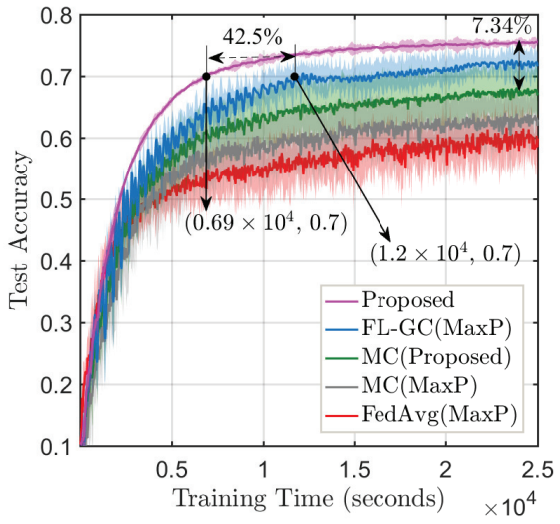
This section evaluates the proposed approach on the CIFAR-10 dataset by training a VGG-11 model [13]. For the dataset splitting, similar to [1], [5], [14], we allocate no more than two data classes to each client, which is a widely adopted data heterogeneous setting in existing works. Unless otherwise specified, the default system settings are listed in Table I.

TABLE I
System Settings

Parameter	Value	Parameter	Value
N	100	C	10
B	1MHz	η	0.05
I	5	H_b	64
N_0	-174dBm/Hz	p_n	30dBm
γ_{th}	0dB	R	35.4MB
$T_{\max}$	30s	f_n	1.3GHz
v	2	C_n	76,395,285

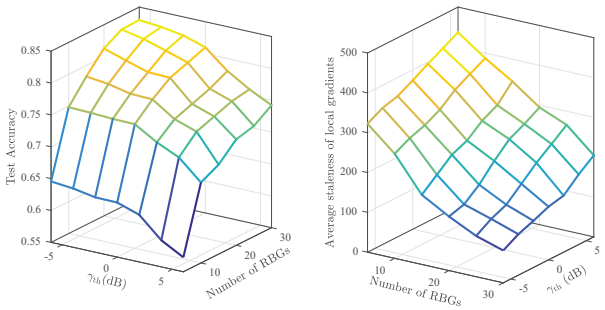
In Fig. 1, we evaluate the proposed approach with the following approaches: 1) FL-GC (Max-P): The ES adopts FL-GC to update the global model, while the client scheduling approach focuses on maximizing the successful transmission client number. 2) MC (Proposed): The ES utilizes the model compensation approach in [8], [9] to update the global model and the proposed client scheduling approach to select clients. 3) MC (Max-P): The global model update method is similar to MC (Proposed), but it maximizes the successful transmission client number in each round. 4) FedAvg (Max-P) [5]: It only aggregates the successfully transmitted local models and maximizes the successful transmission client number. Fig. 1 shows that the proposed approach converges faster and achieves significant accuracy improvement than the benchmarks. It improves 7.34% accuracy than the model compensation approach with the maximum successful client number. In addition, the proposed approach only spends 0.69×10^4 seconds to achieve 70% accuracy, while the MC

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(a)

Fig. 1. Comparison of learning performance of different FL algorithms.



(a) (b)

Fig. 2. Impacts of wireless network parameters on FL performance.

with Max-P approach requires 1.2×10^4 seconds. That is, the proposed is able to save 42.5% time to obtain 70% accuracy. Note that, the proposed approach to minimize the average staleness of local gradients achieves better learning performance than the FL-GC with Max-P approach. This is because that lower average staleness may reduce the gradient estimation error and thus improve the learning accuracy.

Fig. 2 examines how wireless network parameters affect the learning performance of the proposed approach. Note that the accuracy and average staleness in Fig. 2 are obtained after 2.5×10^4 seconds of training. It is observed from Fig. 2(a) that the learning accuracy of the proposed approach increases with the increasing of RBG number and the decreasing of the decoder threshold. The reason can be found in Fig.2(b). When the RBG number is large and the decoder threshold is small, more clients may successfully transmit their gradients to the server. Thus, the convergence bound is narrowed, and the learning accuracy is improved. This further verified our theoretical analysis result which shows that small average staleness of local gradients helps improve learning performance.

V. Conclusion

In this work, we proposed a novel FL-GC framework to mitigate the adverse effects of unreliable channels and limited resources. Our convergence analysis results illustrate that minimizing the average staleness of local gradients helps enhance the FL performance. With this insight, we optimized the client selection and resource allocation strategies for boosting the FL performance. Numerical results demonstrated the effectiveness of the proposed approach.

Appendix

A. *Proof of Lemma* 1

Firstly, we derive the bound of the local model drift from the global model after i iterations as:

$$\begin{aligned}
&\|\mathbf{x}_{n,t}^{(i)} - \mathbf{x}_t\|^2 \\
&= \|\mathbf{x}_{n,t}^{(i-1)} - \eta\tilde{\nabla}\mathcal{L}_n(\mathbf{x}_{n,t}^{(i-1)}) - \mathbf{x}_t\|^2 \\
&= \left\|\mathbf{x}_{n,t}^{(i-1)} - \eta\nabla\mathcal{L}_n(\mathbf{x}_{n,t}^{(i-1)}) - \mathbf{x}_t\right\|^2 \\
&+ \eta^2\left\|\tilde{\nabla}\mathcal{L}_n(\mathbf{x}_{n,t}^{(i-1)}) - \nabla\mathcal{L}_n(\mathbf{x}_{n,t}^{(i-1)})\right\|^2 \\
&\overset{(a)}{\leq} \|\mathbf{x}_{n,t}^{(i-1)} - \eta\nabla\mathcal{L}_n(\mathbf{x}_{n,t}^{(i-1)}) - \mathbf{x}_t\|^2 + \eta^2\sigma^2 \\
&= \|\mathbf{x}_{n,t}^{(i-1)} - \mathbf{x}_t\|^2 + \eta^2\|\nabla\mathcal{L}_n(\mathbf{x}_{n,t}^{(i-1)})\|^2 \\
&- 2\left\langle\frac{1}{\sqrt{I-1}}(\mathbf{x}_{n,t}^{(i-1)} - \mathbf{x}_t), \eta\sqrt{I-1}\nabla\mathcal{L}_n(\mathbf{x}_{n,t}^{(i-1)})\right\rangle + \eta^2\sigma^2 \\
&\leq (1 + \frac{1}{I-1})\|\mathbf{x}_{n,t}^{(i-1)} - \mathbf{x}_t\|^2 \\
&+ I\eta^2\|\nabla\mathcal{L}_n(\mathbf{x}_{n,t}^{(i-1)})\|^2 + \eta^2\sigma^2 \\
&\overset{(b)}{\leq} (1 + \frac{3}{2(I-1)})\|\mathbf{x}_{n,t}^{(i-1)} - \mathbf{x}_t\|^2 + 2I\eta^2G_1^2 + \eta^2\sigma^2, \quad (21)
\end{aligned}$$

where (a) is obtained by adding and subtracting $\nabla\mathcal{L}_n(\mathbf{x}_{n,t}^{(i-1)})$ into $\tilde{\nabla}\mathcal{L}_n(\mathbf{x}_{n,t}^{(i-1)})$, then using Assumption 2 and (b) follows Assumptions 2 and 3. By telescoping (21), we have

$$\begin{aligned}
&\|\mathbf{x}_{n,t}^{(i)} - \mathbf{x}_t\|^2 \\
&\leq (2I\eta^2G_1^2 + \eta^2\sigma^2)\frac{(1 + \frac{3}{2(I-1)})^{I-1} - 1}{\frac{3}{2(I-1)}}. \quad (22)
\end{aligned}$$

In (22), we have $(1 + \frac{3}{2(I-1)})^{I-1} = (1 + \frac{3}{2(I-1)})^{\frac{2(I-1)}{3}\frac{3}{2}} \leq e^{\frac{3}{2}} \leq 5$, and $\frac{2(I-1)}{3} \leq I - 1$. Thus, we have

$$\|\mathbf{x}_{n,t}^{(i)} - \mathbf{x}_t\|^2 \leq 4(I-1)(2I\eta^2G_1^2 + \eta^2\sigma^2). \quad (23)$$

Based on the above analysis, for round t and t' with $t > t'$, we have

$$\|\mathbf{x}_t - \mathbf{x}_{t'}\|^2$$

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$$
\begin{aligned}
&= \Big\| \sum\nolimits_{j=t'}^{t-1} (\mathbf{x}_{j+1} - \mathbf{x}_j) \Big\|^2 \\
&\leq 2\eta^2 \Big\| \sum_{j=t'}^{t-1} \frac{1}{N} \sum_{n=1}^{N} \tilde{\mathbf{g}}_{n,j} \Big\|^2 + 2\eta^2 \Big\| \sum_{j=t'}^{t-1} \frac{1}{N} \sum_{n=1}^{N} \widehat{\mathbf{g}}_{n,j} \Big\|^2 \\
&\leq 2\eta^2 \Big\| \sum\nolimits_{j=t'}^{t-1} \frac{1}{N} \sum_{n=1}^{N} \tilde{\mathbf{g}}_{n,j} \Big\|^2 + 2\eta^2 (t-t')^2 G_2^2, \qquad (24)
\end{aligned}
$$

where the last inequality is due to Assumption 3. Below we bound the first term on the RHS of (24) as

$$
\begin{aligned}
&\Big\| \sum\nolimits_{j=t'}^{t-1} \frac{1}{N} \sum\nolimits_{n=1}^{N} \frac{1}{I} \sum\nolimits_{i=0}^{I-1} \tilde{\nabla} \mathcal{L}_n(\mathbf{x}_{n,j}^{(i)}) \Big\|^2 \\
&\leq (t-t') \sum_{j=t'}^{t-1} \frac{1}{N} \sum_{n=1}^{N} \frac{1}{I} \sum_{i=0}^{I-1} \| \tilde{\nabla} \mathcal{L}_n(\mathbf{x}_{n,j}^{(i)}) \|^2 \\
&\leq 3(t-t') \sum_{j=t'}^{t-1} \frac{1}{N} \sum_{n=1}^{N} \frac{1}{I} \sum_{i=0}^{I-1} \Big(\| \tilde{\nabla} \mathcal{L}_n(\mathbf{x}_{n,j}^{(i)}) - \nabla \mathcal{L}_n(\mathbf{x}_{n,j}^{(i)}) \|^2 \\
&+ \| \nabla \mathcal{L}_n(\mathbf{x}_{n,j}^{(i)}) - \nabla \mathcal{L}_n(\mathbf{x}_j) \|^2 + \| \nabla \mathcal{L}_n(\mathbf{x}_j) \|^2 \Big) \\
&\overset{(a)}{\leq} 3(t-t') \sum_{j=t'}^{t-1} \frac{1}{N} \sum_{n=1}^{N} \frac{1}{I} \sum_{i=0}^{I-1} \Big(\sigma^2 + G_1^2 + L^2 \| \mathbf{x}_{n,j}^{(i)} - \mathbf{x}_j \|^2 \Big) \\
&\overset{(b)}{\leq} 3(t-t')^2 \Big((1 + \frac{I-1}{I^2}) \sigma^2 + \frac{3I-2}{I} G_1^2 \Big), \qquad (25)
\end{aligned}
$$

where (a) is due to Assumptions 1-3, (b) is due to (23). Substituting (25) into (24), we complete the proof.

B. Proof of Theorem 1

Based on Corollary 1 in [4], let Assumptions 1-3 hold, the convergence bound satisfy

$$
\begin{aligned}
&\mathbb{E}[\mathcal{L}(\mathbf{x}_{t+1}) - \mathcal{L}(\mathbf{x}^*)] \\
&\quad \leq (1 - L^2\eta^2)^{T-1} \mathbb{E}[\mathcal{L}(\mathbf{x}_0) - \mathcal{L}(\mathbf{x}^*)] + c \\
&\qquad + \sum_{t=1}^{T-1} (1 - L^2\eta^2)^t L\eta^2 \Big\| \widehat{\mathbf{g}}_t - \sum_{n=1}^{N} \tilde{\mathbf{g}}_{n,t} \Big\|^2, \qquad (26)
\end{aligned}
$$

where $c = \frac{1 - L^2\eta^2 - (1-L^2\eta^2)^T}{L} ((2I^2 - 2I + 1)\sigma^2 + (I-1)^2 G_1^2)$ is a system hyperparameters-related constant. Below we bound the gradient difference $\| \widehat{\mathbf{g}}_t - \sum_{n=1}^{N} \tilde{\mathbf{g}}_{n,t} \|^2$ as

$$
\begin{aligned}
&\mathbb{E} \Big\| \widehat{\mathbf{g}}_t - \sum\nolimits_{n=1}^{N} \tilde{\mathbf{g}}_{n,t} \Big\|^2 \\
&= \mathbb{E} \Big\| \frac{1}{N} \sum\nolimits_{n=1}^{N} (1 - \theta_{n,t} s_{n,t}) (\tilde{\mathbf{g}}_{n,t} - \widehat{\mathbf{g}}_{n,t}) \Big\|^2 \\
&\leq \frac{1}{N} \sum\nolimits_{n=1}^{N} (1 - \theta_{n,t} s_{n,t}) \mathbb{E} \| \tilde{\mathbf{g}}_{n,t} - \widehat{\mathbf{g}}_{n,t} \|^2 \\
&\overset{(a)}{\leq} \frac{1}{N} \sum_{n=1}^{N} (1 - \theta_{n,t} s_{n,t}) \frac{\varepsilon}{2} \mathbb{E} \left\| \mathbf{x}_t - \mathbf{x}_{t - \tau_{n,t}} \right\|^2 \\
&\overset{(b)}{\leq} 3\eta^2 \varepsilon c_1 \mathbb{E} \Big[\frac{1}{N} \sum\nolimits_{n=1}^{N} (1 - \theta_{n,t} s_{n,t}) \tau_{n,t}^2 \Big] \\
&\overset{(c)}{=} 3\eta^2 \varepsilon c_1 \mathbb{E} \Big[\frac{1}{N} \sum_{n=1}^{N} (1 - \theta_{n,t} s_{n,t}) (\tau_{n,t-1} + 1)^2 \Big] \\
&\overset{(c)}{=} \frac{3\eta^2 \varepsilon c_1}{N} \sum\nolimits_{n=1}^{N} (\tau_{k,t-1} + 1)^2 \\
&\quad \times \Big(1 - \theta_{n,t} \sum\nolimits_{c=1}^{C} z_{n,t}^{(c)} \exp\Big(- \frac{\gamma_{\text{th}} B N_0}{p_n d_n^{-v}} \Big) \Big), \qquad (27)
\end{aligned}
$$

where (a) comes from the Taylor's theorem [15], (b) is due to Lemma 1, (c) comes from the evolution of $\tau_{n,t}$ in (14). Substituting (27) into (26), we complete the proof.

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